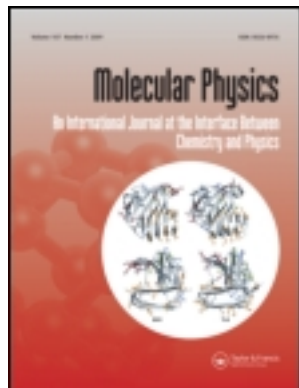


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An analytic structure factor for macroion solutions

John B. Hayter^a & Jeff Penfold^b

^a Institut Laue-Langevin, 156X, 38042, Grenoble Cédex, France

^b Neutron Division, Rutherford and Appleton Laboratories, Chilton, Didcot, Oxon, England

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An analytic structure factor for macroion solutions

by JOHN B. HAYTER

Institut Laue-Langevin, 156X, 38042 Grenoble Cédex, France

and JEFF PENFOLD

Neutron Division, Rutherford and Appleton Laboratories,
Chilton, Didcot, Oxon, England

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We assume the time-averaged structure of a macroion solution to be determined only by the repulsive screened Coulomb pair potential between finite macroions. The counter ions and solvent are treated as a uniform neutralizing background which determines the screening of the potential. We solve the Ornstein-Zernike equation in the mean spherical approximation to obtain closed analytic forms for the direct correlation function $c(r)$ and the structure factor $S(Q)$. In the zero charge limit, the Percus-Yevick hard sphere solution is recovered. As charge is added to the macroions at given volume fraction, the isothermal compressibility decreases and $S(Q)$ shows increasing structure, eventually exhibiting solid-like behaviour. The results provide a useful model basis for studying interacting colloidal systems, and finite ion screened one component plasmas in general.

1. INTRODUCTION

A macroion solution is a solution in which one ionic species (the *macroion*) is large enough to dominate many of the solution properties. In particular X-ray, light or neutron scattering from the solution is effectively determined by the behaviour of the macroions alone. Such solutions are particularly important in colloid and surface chemistry; the charged micelles formed in surfactant solutions offer good examples of macroions, as do many large biological molecules.

For structural calculations, a macroion solution may reasonably be modelled as a one component plasma of finite size ions, the counterions and solvent molecules being treated as a uniform neutralizing background which determines the screening in the system. The weakly attractive van der Waals part of the true colloid (DLVO) potential [1] will have little effect on the time-averaged structure, because the repulsive screened Coulomb potential is generally larger than thermal energies at small interparticle separations, and the attractive forces rarely come into play. We assume, therefore, that the structural properties of a macroion solution are governed only by a finite ion screened Coulomb interparticle pair potential.

For spherical macroions taken as having a hard core, the structure factor $S(Q)$ may then be calculated in closed form by solving the Ornstein-Zernike equation [2] in the mean spherical approximation [3]. We present the results in this paper, and give examples of $S(Q)$ as a function of the several parameters of the system, namely the volume fraction, charge and screening (Q is the momentum transfer).

We note that the present problem is formally equivalent to that of a hard sphere potential with a Yukawa tail, for which an analytic form has been obtained by Waisman [4]. In the latter case, however, the resulting non-linear algebraic equations had to be solved numerically. The present closed analytic solution is particularly useful when $S(Q)$ is needed to fit experimental data, where many evaluations of the function may be needed. We also note that the solution for the unscreened case has been given in closed form by Palmer and Weeks [5]. The general background theory, and in particular the stability of the system, has been the subject of recent reviews [6, 7] and will not be further discussed here.

2. THE DIRECT CORRELATION FUNCTION

The repulsive potential between two identical spherical macroions of diameter σ is [1]

$$U(r) = \pi\epsilon_0\sigma^2 \psi_0^2 \exp[-\kappa(r-\sigma)]/r, \quad r > \sigma, \quad (1)$$

where r is the interionic centre-to-centre distance, ψ_0 is the surface potential, ϵ is the dielectric constant of the solvent medium, and ϵ_0 is the permittivity of free space. Here, κ is the usual Debye-Hückel inverse screening length, determined by the ionic strength of the solution. The surface potential ψ_0 is related to the electronic charge z_m on the macroion to a good approximation by

$$\psi_0 = z_m / \pi\epsilon_0\sigma(2 + \kappa\sigma), \quad (2)$$

where the final term reflects the mutual exclusion by the macroions of counterions from each other's volume.

We shall formulate the problem in dimensionless units, setting

$$x = r/\sigma, \quad k = \kappa\sigma, \quad K = Q\sigma,$$

and writing

$$\beta U(x) = \gamma \exp(-kx)/x, \quad x > 1 \quad (3 a)$$

where $\beta = 1/k_B T$ is the inverse temperature in energy units, and $\gamma \exp(-k) = \beta\pi\epsilon_0\sigma\psi_0^2$ is the contact potential for a macroion pair, expressed in units of $k_B T$. This potential is valid for colloid systems provided $k \lesssim 6$. There is no theoretical restriction on k in what follows, however, and for general studies of one component plasmas any value may be used.

We now complete the description of the pair potential by taking the macroion to have a hard core, so that

$$\beta U(x) = \infty, \quad x < 1. \quad (3 b)$$

The direct correlation function $c(x)$ may then be obtained for $x < 1$ by solving the Ornstein-Zernike equation

$$h(x) = c(x) + n\sigma^3 \int h(|\mathbf{x} - \mathbf{y}|)c(y) d\mathbf{y} \quad (4)$$

in the mean spherical approximation (MSA), which sets the closure relations [3]

$$c(x) = -\beta U(x), \quad x > 1, \quad (5 a)$$

$$h(x) = -1, \quad x < 1. \quad (5 b)$$

Here, n is the particle number density, $h(x)=g(x)-1$ is the total correlation function, and $g(x)$ is the radial distribution function. Note that in the case of $\gamma \rightarrow 0$ or $k \rightarrow \infty$, (4) and (5) yield the Percus–Yevick hard-sphere result [8, 9]; the unscreened case $k=0$ was solved by Palmer and Weeks [5].

Equations (4) and (5) were solved for $c(x)$ in the range $x < 1$ using the methods of Wertheim [10] and of Waisman and Lebowitz [11] as applied in particular by Palmer and Weeks. We have closely followed the exposition of the latter authors, and the interested reader is referred to their paper [5] for details. The result is

$$c(x) = A + Bx + \frac{1}{2}\eta Ax^3 + \frac{C \sinh kx}{x} + \frac{F (\cosh kx - 1)}{x}, \quad x < 1$$

$$= -\gamma \exp(-kx)/x, \quad x > 1 \quad (6)$$

where $\eta = \pi n \sigma^3 / 6$ is the volume fraction. The constants are given by

$$\left. \begin{aligned} C &= -(\omega_{16}F^2 + \omega_{15}F + \omega_{14})/(\omega_{13}F + \omega_{12}), \\ B &= b_1 + b_2C + b_3F, \\ A &= a_1 + a_2C + a_3F, \end{aligned} \right\} \quad (7)$$

where F is a root of

$$w_4F^4 + w_3F^3 + w_2F^2 + w_1F + w_0 = 0. \quad (8)$$

Other parameters of the problem are

$$V = H(0) = (v_1 + v_2C + v_3F)/A, \quad (9)$$

$$U = H(k) = (k^3 C - 24\eta\gamma)/24\eta\gamma k^2, \quad (10)$$

where

$$H(s) = \int_0^\infty xh(x) \exp(-sx) dx. \quad (11)$$

The coefficients are given in the Appendix. The choice of the correct value of F is direct. All real roots of (8) are extracted by any standard method (see e.g. [12]), and the structure factor $S(K)$ is calculated in a limited K range for each root (see next section). The corresponding radial distribution functions are then calculated from

$$g(x) = 1 + \frac{1}{12\pi\eta x} \int_0^\infty [S(K) - 1] K \sin Kx dK \quad (12)$$

and the root which yields $g(x)=0$ inside the particle is retained as the correct physical solution (condition (5 b)). The procedure is straightforward to automate [13, 14].

3. THE STRUCTURE FACTOR

The structure factor $S(K)$ is obtained by Fourier transforming (4) using (6). The result is

$$S(K) = 1/[1 - 24\eta a(K)], \quad (13)$$

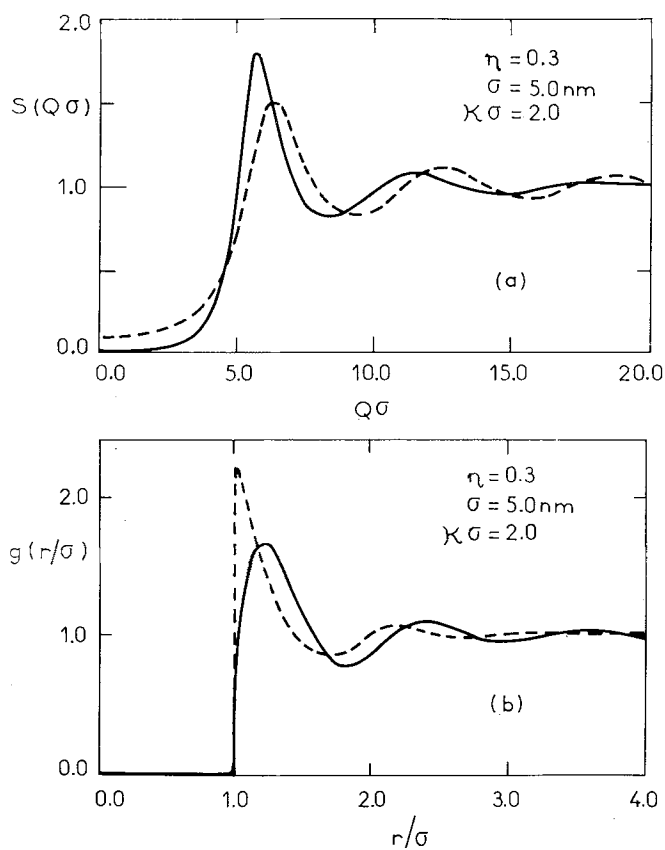


Figure 1. (a) $S(Q\sigma)$ versus $Q\sigma$ for parameters typically encountered in a concentrated macroion solution ($\eta = 0.3$). — $z = 20$ ($\psi_0 = 72 \text{ mV}$); --- $z = 0$ (hard sphere). (b) The corresponding radial distribution functions.

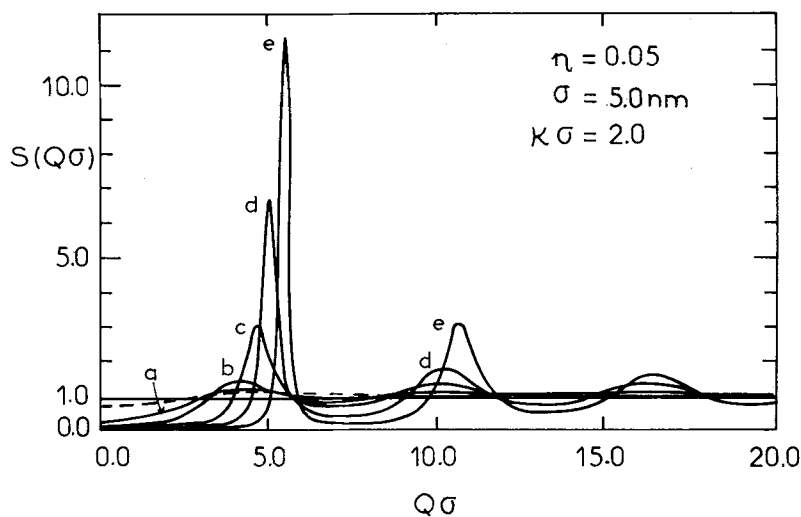


Figure 2. $S(Q\sigma)$ versus $Q\sigma$ as a function of macroion charge for a dilute macroion solution ($\eta = 0.05$). (a) $z = 10$, (b) $z = 20$, (c) $z = 50$, (d) $z = 100$, (e) $z = 200$; --- $z = 0$ (hard sphere).

where $K = Q\sigma$ and

$$\begin{aligned}
 a(K) = & A(\sin K - K \cos K)/K^3 \\
 & + B[(2/K^2 - 1)K \cos K + 2 \sin K - 2/K]/K^3 \\
 & + \eta A[24/K^3 + 4(1 - 6/K^2) \sin K \\
 & - (1 - 12/K^2 + 24/K^4)K \cos K]/2K^3 \\
 & + C(k \cosh k \sin K - K \sinh k \cos K)/K(K^2 + k^2) \\
 & + F[k \sinh k \sin K - K(\cosh k \cos K - 1)]/K(K^2 + k^2) \\
 & + F(\cos K - 1)/K^2 \\
 & - \gamma \exp(-k)(k \sin K + K \cos K)/K(K^2 + k^2).
 \end{aligned} \tag{14}$$

The long wavelength limit is given by

$$S(0) = -1/A. \tag{15}$$

As a slight digression, the point ion limit is also of interest. Writing the potential in terms of the charge z_m on the ion and letting $\sigma \rightarrow 0$ yields

$$S(Q) \rightarrow 1/[1 + \kappa_m^2/(Q^2 + \kappa^2)], \tag{16}$$

where we have defined (by analogy with the inverse screening length κ)

$$\kappa_m^2 = \beta n z_m^2 / \epsilon_0 \epsilon. \tag{17}$$

Expanding (16) at low density gives

$$S(Q) \rightarrow 1 - \kappa_m^2/(Q^2 + \kappa^2), \tag{18}$$

which is the Debye-Hückel result. Note that in general, however, the MSA fails at low density; letting $n \rightarrow 0$ in (4) and using (5a) yields $g(x) \rightarrow 1 - \beta U(x)$ for $x > 1$. Since $U(x)$ is generally larger than thermal energies for small interparticle separations, $g(x)$ will generally be negative (and hence unphysical) near the particle at very low densities. This does not present a problem for many colloid studies of current interest, where volume fractions are generally greater than one per cent.

Calculated structure factors for parameters typically encountered in aqueous macroion solutions ($T = 298$ K, $\epsilon = 78.3$) are shown in figures 1 and 2, with the corresponding hard sphere (zero charge) results shown for comparison. The charge $z = z_m/e^-$ is expressed in electrons/macroion, and in all figures we have taken the diameter $\sigma = 5$ nm.

The effect of adding charge to the system is readily apparent. Initially, there is a decrease in the isothermal compressibility, as reflected in $S(0)$. This is followed by an increasing build-up of structure which eventually becomes lattice-like (curve (e) of figure 2). The fundamental qualitative difference between the charged and uncharged systems is evident in the radial distribution function, figure 1 (b); the charged macroions sit further apart and avoid contact, as expected.

Figures 3 to 5 survey the properties of the structure factor near its first maximum, as a function of volume fraction, surface potential and screening. In figure 4, it is interesting to note that the height of the first peak passes through

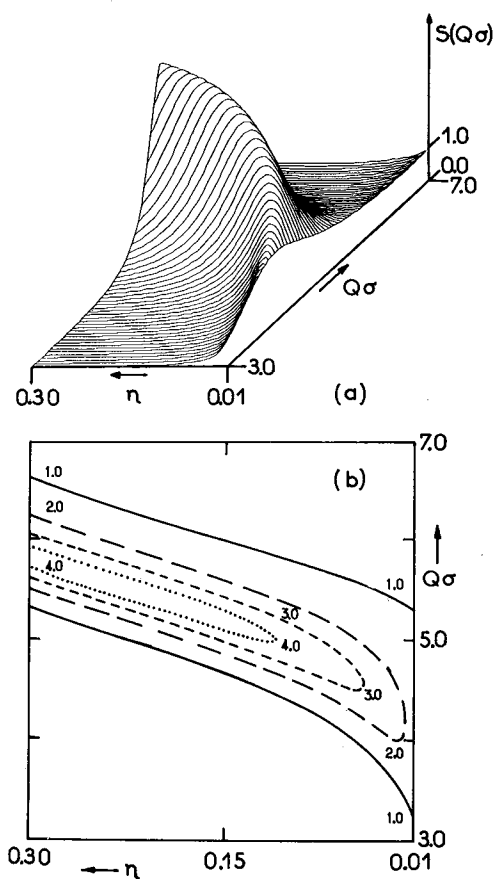


Figure 3. The structure factor near the first maximum as a function of volume fraction at fixed surface potential and screening. ($\psi_0 = 180$ mV, $k = 2$, corresponding to $z = 50$.) (a) Isometric plot, (b) associated contour plot.

a maximum at $k \sim 5$. As the screening constant increases from zero, the first effect is to produce an increase in structure due to the larger first derivative of the potential. This must eventually be offset by the fact that as $k \rightarrow \infty$ the hard sphere result is again obtained, and the latter shows little structure at the concentration used to calculate the figure ($\eta = 0.05$).

We believe that the present closed-form structure factor will prove useful in interpreting scattering data from many interacting colloidal systems of current interest.

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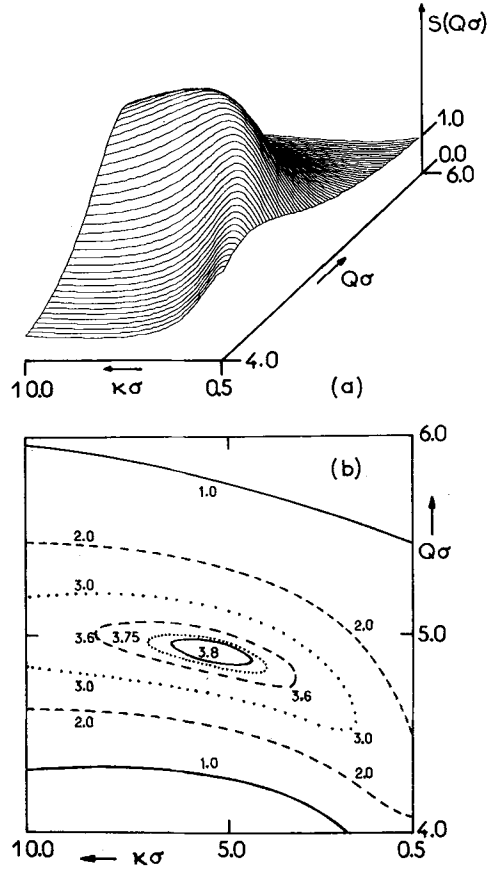


Figure 4. The structure factor near the first maximum as a function of the screening at fixed surface potential and volume fraction. ($\psi_0 = 180$ mV, $\eta = 0.05$).

APPENDIX

We present here expressions for the coefficients appearing in (7), (8) and (9).
Set

$$\begin{aligned}
 \delta &= 1 - \eta, \\
 \alpha_1 &= -(2\eta + 1)\delta/k, \\
 \alpha_2 &= (14\eta^2 - 4\eta - 1)/k^2, \\
 \alpha_3 &= 36\eta^2/k^4, \\
 \beta_1 &= -(\eta^2 + 7\eta + 1)\delta/k, \\
 \beta_2 &= 9\eta(\eta^2 + 4\eta - 2)/k^2, \\
 \beta_3 &= 12\eta(2\eta^2 + 8\eta - 1)/k^4, \\
 \nu_1 &= -(\eta^3 + 3\eta^2 + 45\eta + 5)\delta/k, \\
 \nu_2 &= (2\eta^3 + 3\eta^2 + 42\eta - 20)/k^2,
 \end{aligned}$$

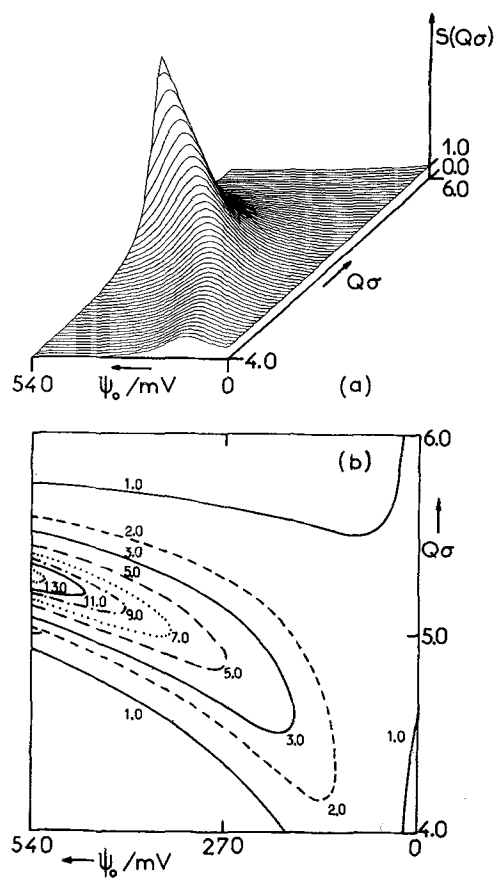


Figure 5. The structure factor near the first maximum as a function of the surface potential at fixed screening and volume fraction. ($k=2$, $\eta=0.05$). The maximum surface potential corresponds to $z=150$.

$$\nu_3 = (2\eta^3 + 30\eta - 5)/k^4,$$

$$\nu_4 = \nu_1 + 24\eta k \nu_3,$$

$$\nu_5 = 6\eta(\nu_2 + 4\nu_3),$$

$$\phi_1 = 6\eta/k,$$

$$\phi_2 = \delta - 12\eta/k^2,$$

$$\tau_1 = (\eta + 5)/5k,$$

$$\tau_2 = (\eta + 2)/k^2,$$

$$\tau_3 = -12\eta\gamma \exp(-k)(\tau_1 + \tau_2),$$

$$\tau_4 = 3\eta k^2(\tau_1^2 - \tau_2^2),$$

$$\tau_5 = 3\eta(\eta + 8)/10 - 2(2\eta + 1)^2/k^2.$$

Then

$$\begin{aligned}
 a_1 &= \{24\eta\gamma \exp(-k)[\alpha_1 + \alpha_2 + (1+k)\alpha_3] - (2\eta+1)^2\}/\delta^4, \\
 a_2 &= 24\eta[\alpha_3(\sinh k - k \cosh k) + \alpha_2 \sinh k - \alpha_1 \cosh k]/\delta^4, \\
 a_3 &= 24\eta[(2\eta+1)^2/k^2 - \delta^2/2 + \alpha_3(\cosh k - 1 - k \sinh k) \\
 &\quad - \alpha_1 \sinh k + \alpha_2 \cosh k]/\delta^4, \\
 b_1 &= \{3\eta(\eta+2)^2/2 - 12\eta\gamma \exp(-k)[\beta_1 + \beta_2 + (1+k)\beta_3]\}/\delta^4, \\
 b_2 &= 12\eta[\beta_3(k \cosh k - \sinh k) - \beta_2 \sinh k + \beta_1 \cosh k]/\delta^4, \\
 b_3 &= 12\eta[\delta^2(\eta+2)/2 - 3\eta(\eta+2)^2/k^2 - \beta_3(\cosh k - 1 - k \sinh k) \\
 &\quad + \beta_1 \sinh k - \beta_2 \cosh k]/\delta^4, \\
 v_1 &= [(2\eta+1)(\eta^2 - 2\eta + 10)/4 - \gamma \exp(-k)(\nu_4 + \nu_5)]/5\delta^4, \\
 v_2 &= (\nu_4 \cosh k - \nu_5 \sinh k)/5\delta^4, \\
 v_3 &= [(\eta^3 - 6\eta^2 + 5)\delta - 6\eta(2\eta^3 - 3\eta^2 + 18\eta + 10)/k^2 \\
 &\quad + 24\eta\nu_3 + \nu_4 \sinh k - \nu_5 \cosh k]/5\delta^4.
 \end{aligned}$$

Define

$$\begin{aligned}
 p_1 &= [\gamma \exp(-k)(\phi_1 - \phi_2)^2 - (\eta+2)/2]/\delta^2, \\
 p_2 &= [(\phi_1^2 + \phi_2^2)\sinh k + 2\phi_1\phi_2 \cosh k]/\delta^2, \\
 p_3 &= [(\phi_1^2 + \phi_2^2)\cosh k + 2\phi_1\phi_2 \sinh k + \phi_1^2 - \phi_2^2]/\delta^2, \\
 t_1 &= \tau_3 + \tau_4 a_1 + \tau_5 b_1, \\
 t_2 &= \tau_4 a_2 + \tau_5 b_2 + 12\eta(\tau_1 \cosh k - \tau_2 \sinh k), \\
 t_3 &= \tau_4 a_3 + \tau_5 b_3 + 12\eta[\tau_1 \sinh k - \tau_2(\cosh k - 1)] \\
 &\quad - 2\eta(\eta+10)/5 - 1, \\
 \mu_1 &= t_2 a_2 - 12\eta v_2^2, \\
 \mu_2 &= t_1 a_2 + t_2 a_1 - 24\eta v_1 v_2, \\
 \mu_3 &= t_2 a_3 + t_3 a_2 - 24\eta v_2 v_3, \\
 \mu_4 &= t_1 a_1 - 12\eta v_1^2, \\
 \mu_5 &= t_1 a_3 + t_3 a_1 - 24\eta v_1 v_3, \\
 \mu_6 &= t_3 a_3 - 12\eta v_3^2, \\
 \lambda_1 &= 12\eta p_2^2, \\
 \lambda_2 &= 24\eta p_1 p_2 - 2b_2, \\
 \lambda_3 &= 24\eta p_2 p_3, \\
 \lambda_4 &= 12\eta p_1^2 - 2b_1, \\
 \lambda_5 &= 24\eta p_1 p_3 - 2b_3 - k^2, \\
 \lambda_6 &= 12\eta p_3^2.
 \end{aligned}$$

Then the coefficients of the quartic are

$$w_4 = \omega_{16}^2 - \omega_{13}\omega_{36},$$

$$w_3 = 2\omega_{16}\omega_{15} - \omega_{13}(\omega_{35} + \omega_{26}) - \omega_{12}\omega_{36},$$

$$w_2 = \omega_{15}^2 + 2\omega_{16}\omega_{14} - \omega_{13}(\omega_{34} + \omega_{25}) - \omega_{12}(\omega_{35} + \omega_{26}),$$

$$w_1 = 2\omega_{15}\omega_{14} - \omega_{13}\omega_{24} - \omega_{12}(\omega_{34} + \omega_{25}),$$

$$w_0 = \omega_{14}^2 - \omega_{12}\omega_{24},$$

where $\omega_{ij} = \mu_i\lambda_j - \mu_j\lambda_i$. We also have by direct calculation $\lambda_3^2 = 4\lambda_1\lambda_6$, and hence $w_4 = (\mu_1\lambda_6 + \lambda_1\mu_6 - \lambda_3\mu_3/2)^2 \geq 0$.

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